

Project Proposal

Analysis and Presentation of Construction Data possibly with AI - Ex35

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Course: COMP702 / COMP708

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Version	Date	Author(s)	Notes
1.0	23/03/2026	Becky Zhou	This broadly based on BCIS project assessment guidelines

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Executive Summary

We are undertaking this project to help YourQS, a quantitative surveying company that uses a databased driven system to give cost estimates. This is implemented through building 3D models to help architects and clients to gain the precise cost and extract meaningful data for decision making.

Problem:

- Large Quantity of project data is available but no scope for gaining meaningful insights.
- Lack of ability to gain averages, trends, estimates etc:
- No ability to compare Historical data with new projects
- Lack of ability to gain insights of “what if” questions of clients/architects
- No feature of AI present to make cost estimation feasible.

Solution:

- Developing a data focused system where insights can be obtained.
- Using SQL, Python and other relevant skills to integrate database with analytical system.
- Creating a simple interface prototype to illustrate the functionality.
- Integrate AI to give users a better understanding of their questions and provide further details to help architects/clients with decision making.

Deliverables:

- Data analysis output including trends, averages and comparison between data values.
- Users able to perform “What if” analysis such as “What if the floor was concrete instead of timber”
- Develop a methodology to compare the costs in one model with averages from similar past model
- AI having the capability to offer a plain language querying of the costing data.
- Building a dashboard to show the functionality of the project as the UI design is not needed by the client.

Project Scope and Timeline

- Total duration: 2 Semesters
- Project Proposal (**Part 1**) due 29th March (Week 5) and Presentation in Week 6.
- Midterm Project review (Part 1) due 31st May
- **Part 2** Client Feedback Form (Week 10)
- Reflective report and Team Portfolio (Week 13)
- Poster submission and showcase (Week 14)

Estimate Cost:

- Total estimated Cost is coming to \$61,375.80
- Major cost includes Team labour and mentor hours. (Details are justified further in report)

Risks:

- Integrating data analysis skills with the existing Firebird Database.
- AI limitation for the functionality until we get the approval from client.

1. Terms of Reference

This project aims to develop a system that improves the analysis and presentation of construction cost data for residential builds. The focus is on enabling clear comparison of different materials and finishes to support more informed decision-making.

The project will involve extracting and processing construction data, particularly related to material costs, and applying analytical methods to identify differences and trends. The results will be presented through structured visualisations and summaries, allowing users to evaluate multiple options efficiently within a single system. There is also potential to explore AI-based querying to enhance user interaction, depending on feasibility.

The scope is limited to working with existing data and available tools, without modifying the underlying database structure. The outcome will be a functional solution supported by project documentation, demonstrating improved data analysis and comparison capabilities.

2. Rationale for the Project

Material selection plays a significant role in determining the overall cost of residential construction projects. However, effectively comparing different material and finish options can be challenging due to the way the current software is designed.

This project aims to improve the usability of construction data by enabling more efficient comparison of material costs. By presenting multiple options in a clear and structured format, the system will support identifying cost-effective solutions while maintaining design requirements.

Improved data visualisation and analysis will also reduce the time required to interpret information and support more confident decision-making. Overall, the project is justified by its potential to enhance efficiency, improve data accessibility, and provide greater value from existing construction data.

3. Project Objective and Scope

3.1 Project Objective

The objective of this project is to develop a prototype web-based platform that enables the analysis and visualisation of construction data provided by Your QS. The system will demonstrate how data analytics can support cost comparison, trend analysis, and decision making. It will transform raw data into meaningful insights through interactive dashboards, with optional exploration of natural language querying.

3.2 High-Level Requirements

The system will provide an end-to-end solution for retrieving, analysing, and visualising construction data through a web-based interface. It will include core functionality such as data filtering, cost comparison, and visualisation, while also meeting key non-functional requirements including performance, usability, and data security.

3.2.1 Privacy Considerations

The system will ensure that all construction and cost data is handled securely and remains protected from unauthorised access. Data will only be accessed through approved methods, and any sensitive information will not be exposed or shared externally. Appropriate measures will be taken to maintain data privacy, especially when integrating with external tools or services.

3.3 Key Stakeholders

Key stakeholders include the client (YourQS), who provides the data and project requirements, the project team responsible for development and delivery, and end users such as architects, estimators, and builders who will interact with the system. Academic supervisors also act as stakeholders by providing guidance and assessment.

3.4 Scope-Related Risks

The project may face risks related to technical integration, time constraints, and potential expansion of scope beyond initial requirements. These risks will be managed by prioritising core features, maintaining clear project boundaries, and following a structured development plan.

3.5 Technical Requirements

The system will be developed as a web-based application using a backend framework (e.g. FastAPI) and data processing tools. It will integrate with the YourQS Firebird database via an API layer and provide interactive data visualisation through a user-friendly interface. The system must ensure reliable performance, scalability, and secure data handling.

4. Skills Analysis

4.1 Team Skills Register

The team has a strong foundation in Python, SQL and data analysis, which aligns well with the analytical goals of the project.

However, there's limited experience with C# (.NET) and Blazor which are required for integration with the client's environment. These gaps will be addressed through a structured upskilling plan.

4.2 Missing Skills and Upskilling Plan

The following key skill gaps have been identified:

- Limited experience with C# and .NET development
- Limited experience with Blazor frontend development
- Limited experience with AI integration in analytics systems

Upskilling Strategy:

C# / .NET:

The team will develop basic C# skills required for backend integration. The focus will be on understanding API endpoints, request handling and communication with other services.

Blazor Frontend:

The team will acquire foundational knowledge of Blazor to support the development of a simple web-based interface. The focus will be on building basic interactive components and integrating them with backend.

AI Integration:

The team will explore AI integration as an assistive layer for interpreting analytical results, ensuring that sensitive data isn't directly exposed to external systems (client's request).

Learning Resources:

Will be used a combination of:

- Official documentation (Microsoft .NET, ASP.NET)
- Online tutorial and platforms (YouTube, technical blogs)
- Example code and peer learning within the team

Time Allocation:

Time for skills development will be allocated during the early stages of the project. Initial weeks will include focused learning and experimentation to ensure the team can effectively implement the required technology stack without delaying core development (approx. 2 weeks).

5. Team Roles and Responsibilities

To ensure efficient development and clear accountability, the team will adopt a role structure aligned with the system architecture. This approach allows parallel development across different components, including the backend, analytics part and frontend interface.

Roles:

Becky Zhou - Project Manager / Backend Developer (C# / .NET)

Becky is studying Computer Science, majoring in Software Development and Data Science. She is also the team leader for this project. She is responsible for overall project coordination, including planning, scheduling, and tracking progress. She ensures that tasks are completed on time and that the team's work aligns with the project goals.

In addition, she is responsible for backend development using C#/.NET, including implementing API endpoints and managing system integration between different components.

Max Makarov - Data Engineer

Max is studying Data Science with a minor in Finance. He is responsible for managing data flow between the database, backend, and analytics components. This includes data extraction, transformation, and ensuring smooth integration across the system. He also makes sure cost data is accurate and reliable for analysis and supports the development of analytical logic and insights used within the system. He works with database structures and relationships, ensuring the data is well-organised and optimised for analytical use.

Mantej Singh - Data Analyst

Mantej is studying Data Science with a focus on Network and Cybersecurity.

He is responsible for data processing and analytical logic using Python and SQL. This includes data preprocessing, cleaning, aggregation, EDA (Exploratory data analysis) and preparing structured outputs for visualisation and AI processing.

In addition, he is responsible for ensuring data security and safety, including handling sensitive data appropriately and ensuring that no data leakage occurs when interacting with AI components.

Jackson Full - Frontend Developer (Blazor)

Jackson is studying Data Science with a minor in Business Management. He is responsible for developing the user interface using Blazor. This includes visualisations, dashboards, interactive components that allow users to query and explore the data. He also considers business needs when designing the interface, ensuring that the system supports practical decision-making and provides clear insights for users such as architects and clients.

Note:

Even though there're defined tasks to be handled by the team, there will be flexibility to ensure that team members can contribute to different areas. This ensures flexibility and eliminates any possible bottlenecks during development process.

6. Project Feasibility

The project is considered feasible within the available timeframe due to a modular architecture that separates system into a C# based backend, a Blazor based frontend, and a Python analytics layer. This allows parallel development and reduces overall complexity.

The system will use C#, Blazor, Python and SQL, with MS Access as the initial data source, supported by tools such as GitHub, OneDrive, Lucidchart, and communication platforms.

The team has strong Python and data analysis skills, but limited experience with C#, system integration and Blazor. These gaps will be addressed through targeted upskilling task allocation.

Alternative approaches including fully Python-based or fully C#-based platforms were considered. A hybrid approach was selected to balance client requirements and analytical efficiency.

7. Team Schedule for Part 1

Below is a high- level weekly plan for Part 1 till the end of this semester, reflecting 12-15 hours of commitment weekly from the team both physically and individually towards the Project.

7.1 Weekly Offsite and Co-location Schedule weekly

Activity	Day	Time	Hours	Purpose
Team Meeting WG building (Hybrid)	Tuesday	12pm-3pm	3	Planning and Progress Update
Co- location R&D lab WZ	Thursday	10am-2pm	4	Collaborative group work
Off-site Work	Monday	2pm-5pm	3	Individual work
Off-Site work	Saturday	11am-1pm	2	Individual Work
Off-site work	Friday	4pm – 6pm	2	Individual work

7.2 Supervisor Meeting Schedule weekly

Type	Day	Time	Duration	Mode	Purpose
Mentor meeting	Wednesday (or any day mentor is available)	5pm -6pm	1 hour	Online	Progress feedback and questions

7.3 Client Meeting Schedule

Type	Day	Time	Duration	Mode	Purpose
Client meeting	Thursday or (any other day depending on availability.)	11am – 12pm	1 hour	Physical	Client requirements and Feedback

8. Risk and Issues Management

8.1 Risk Management Process

The project uses a structured risk management process to identify, assess, and manage risks throughout development. Risks such as database integration, system performance, and data accuracy are evaluated based on their likelihood and impact. Mitigation strategies, including early testing, data validation, and incremental development, are used to reduce risk. Risks are continuously monitored and updated to ensure they are effectively managed.

8.2 Risk Register Summary

A risk register is used to track and prioritise project risks. The highest priority risk is difficulty integrating with the Firebird database, followed by time constraints and performance issues when working with large datasets. Other key risks include limited experience with data analytics tools, incorrect interpretation of data, and data security concerns. Additional risks include communication issues, technical development challenges, difficulties in creating effective visualisations, and changes in material costs affecting data accuracy. Each risk is prioritised and managed using appropriate mitigation strategies, and the register is regularly updated throughout the project.

8.3 Issue Management Process

Issues are recorded and tracked using a structured system. These may include bugs, data errors, or integration problems identified during development. Issues are prioritised based on impact, with critical issues resolved first. All solutions are tested, and regular reviews ensure issues are addressed efficiently.

9. Project Management Methodology

9.1 Chosen Methodology and Justification

This project uses an Agile Scrum methodology combined with a Data Science approach. Agile Scrum is suitable because the project involves working with a large dataset where the requirements may change as we explore the data and receive feedback from the client. Research also shows that iterative methods like Scrum are useful for data analytics projects, especially when the outcome is developed over time (Baijens et al., 2020). The client also prefers working in 2-week sprints, which makes Scrum an appropriate choice. This allows the team to build the dashboard step by step, receive regular feedback, and continuously improve the system throughout the project. It also helps the team stay flexible and adapt to any changes in requirements.

- Work will be divided into 2-week sprints
- Tasks will be planned at the beginning of each sprint
- Progress will be tracked using a shared tool such as Trello
- The team will have regular meetings, at least once a week, to discuss progress and issues
- Client feedback will be used to guide the next steps

A Data Science approach is also used because the project focuses on analysing construction data from the YourQS database. This involves steps such as understanding the problem, exploring the dataset, preparing the data, analysing it, and evaluating the results (Baijens et al., 2020).

These steps ensure the analysis is structured and produces meaningful insights, such as identifying cost per square metre trends and comparing costs across similar projects. This is important for this project because the YourQS dataset contains historical construction cost data, which can be used to compare average costs and find patterns between similar projects.

Overall, this combination works well because Agile supports flexibility and continuous improvement, while the Data Science approach provides a clear structure for analysis and decision-making.

9.2 Work Breakdown Structure (WBS)

The Work Breakdown Structure (WBS) is used to organise the project into smaller and manageable tasks. This helps the team clearly understand what needs to be done and track progress throughout the project.

The project is divided into the following main phases:

Planning: project setup, team organisation, and proposal preparation

Analysis: understanding client requirements and exploring the dataset

Design: designing the system structure and user interface of the web-based tool

Development: building the data analysis functions and web application

Testing: testing system functionality and validating data outputs

Deployment: finalising and preparing the system for delivery

Each phase is further broken down into smaller tasks and assigned to team members. This structure ensures that the work is distributed clearly and completed in a logical order.

The detailed Work Breakdown Structure (WBS) is provided in the Appendix D.

10. Project Plan

The project plan shows the timeline for the project over 26 weeks across two semesters. It explains how the work will be organised over time, including the main phases and key milestones.

This plan helps the team stay on track, manage time effectively, and make sure all deliverables are completed on time. It also supports the Agile Scrum approach by allowing work to be done in stages with regular improvements.

10.1 Milestone Schedule

The milestone schedule outlines the key stages and important deliverables of the project, including all required assessment tasks.

These milestones act as checkpoints to track progress, identify issues early, and ensure the project stays on schedule.

A detailed Milestone Report is provided in the Appendix E.

10.2 Project Timeline and Phases

The project is divided into the following phases:

Planning (Week 1–5): Project initiation, scope definition, and proposal preparation

Learning (Week 6–13): Upskilling, tool setup, and preparation for development

Analysis (Week 14–17): Data exploration and requirements analysis

Design (Week 17–20): System architecture and dashboard design

Implementation (Week 20–27): Development of the web-based tool and features

Testing (Week 26–29): System testing and improvements

Deployment (Week 29–34): Final system delivery, documentation, and project completion

The timeline includes the two semester breaks (each two weeks).

Work will continue during these periods, allowing the team to maintain progress and stay on schedule.

This timeline supports the Agile methodology by allowing the team to work in iterative cycles, receive feedback, and make continuous improvements while ensuring that key milestones and deliverables are completed on time.

The full project schedule is shown in the Gantt chart in Appendix N.

11. Estimated Project Costs

The total estimated cost for this project is approximately coming to \$61,375.80.
The main costs are driven by team labour and mentor hours.

Along with this a contingency reserve of 10% is also included to account for any unforeseen expenses that may occur during the project.

Other expenses such as Miscellaneous cost, Hardware, Software and Training/Support are also included in the cost estimate. The detailed table and justification are present in the Appendix P alongside references are also present.

References

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Appendix

Appendix A: Standard Disclaimer

**Auckland University of Technology
Bachelor of Computer & Information Sciences
Research & Development Project**

Disclaimer:

Clients should note the general basis upon which the Auckland University of Technology undertakes its student projects on behalf of external sponsors:

While all due care and diligence will be expected to be taken by the students, (acting in software development, research or other IT professional capacities), and the Auckland University of Technology, and student efforts will be supervised by experienced AUT lecturers, it must be recognised that these projects are undertaken in the course of student instruction. There is therefore no guarantee that students will succeed in their efforts.

This inherently means that the client assumes a degree of risk. This is part of an arrangement, which is intended to be of mutual benefit. On completion of the project it is hoped that the client will receive a professionally documented and soundly constructed working software application, some part thereof, or other appropriate set of IT artefacts, while the students are exposed to live external environments and problems, in a realistic project and customer context.

In consequence of the above, the students, acting in their assigned professional capacities and the Auckland University of Technology, disclaim responsibility and offer no warranty in respect of the “technology solution” or services delivered, (e.g. a “software application” and its associated documentation), both in relation to their use and results from their use.

Appendix B: Project Charter

Project Title: Analysis and Presentation of Construction Data possibly with AI - Ex35

Project Start Date: Week 1 Semester 1 2026

Projected Finish: Week 14 Semester 2 2026

Date: 28/03/2026

Budget Information:

Project Manager: Yanbin Liu, yanbin.liu@aut.ac.nz, 0278886666

Project Objectives:

- Analyze and extract meaningful historical construction costs provided by the company.
- Able to identify statistics such as trends, averages, and cost variation across different project families.
- Able to answer “What if” questions by the architects or clients
- Able to support decision making through analysis and AI.

Main Project Success Criteria:

- Accurate and precise information to be extracted through implementation of client questions
- Successful working prototype demonstrating any insights wanted by architects.
- Positive and successful feedback from the client YourQS and mentor.
- Finish all tasks and academic requirements on time with quality.

Approach:

In order for project feasibility and group convenience we will be using a hybrid approach when answering client questions. This is because since the team has more experience in software such as python compared to C#, the hybrid approach of using multiple software's such as C#, Blazor, Python and SQL, with MS will be used.

This approach will allow more effectiveness within the process of project by allowing different group members to focus and upskill in different parts of the project.

The key steps include collecting data, data analysis, Prototype development, audit update, gaining feedback from client and consistently improving over the course of project at every step.

Roles and Responsibilities

The roles are all evenly spread across the four team members as each person in the team has their strengths and weaknesses. Therefore, each member has been assigned a role below to ensure accountability for future communication during the project.

Role	Name	Organization/ Position	Contact Information
Project Manager/ Backend Developer	Becky Zhou	AUT Student	wgk3415@autuni.ac.nz
Data Analyst	Mantej Singh,	AUT Student	Ymh6000@autuni.ac.nz
Frontend developer	Jackson Full	AUT Student	rgt1675@autuni.ac.nz
Data Engineer	Max Makarov	AUT Student	nqy7153@autuni.ac.nz
Mentor	Yanbin Liu	Senior Lecturer	yanbin.liu@aut.ac.nz ,
Project Sponsor	Nick Clements	YourQS	nick@yourqs.co.nz

Sign-off: (Signatures of all above stakeholders. Can sign by their names in table above.)

Appendix C: Scope Statement Document

Project Title: Analysis and Presentation of Construction Data possibly with AI - Ex35
Date: 23/03/26 Prepared by: Jackson Full
Project Justification: This project is justified by the need for YourQS to better utilise its large volume of construction and cost data, which is currently underused for analytical insights. While the existing system supports cost estimation, it lacks capabilities for trend analysis, comparison, and “what-if” scenario evaluation. This project aims to address these gaps by developing a prototype data analytics platform that integrates with the existing YourQS system. By combining C#/.NET backend integration, Python-based analytics, and a Blazor web interface, the project demonstrates how modern technologies can transform raw construction data into meaningful insights. Additionally, the exploration of AI-based natural language querying aims to further enhance usability and decision-making for architects, estimators, and clients.
Product Characteristics and Requirements: 1. The system will be a prototype browser-based platform built using Blazor (frontend), enabling users to interact with construction data through an intuitive interface. 2. It will integrate with the YourQS Firebird/Microsoft Access database via a C#/.NET backend and API layer, ensuring compatibility with the client’s existing environment. 3. The platform will include a Python-based analytics layer (using tools such as Pandas) for data processing, cleaning, aggregation, and analysis. 4. The system will provide interactive dashboards and visualisations (charts, graphs, tables) to present trends, comparisons, and insights clearly.
Summary of Project Deliverables Project management-related deliverables: Business case, charter, team contract, scope statement, WBS, schedule, cost baseline, status reports, final project presentation, final project report, lessons-learned report, and any other required documentation. Product-related deliverables: 1. A functional prototype web-based dashboard developed using Blazor to demonstrate system capabilities. 2. A C#/.NET backend with API endpoints to handle data retrieval and communication between system components. 3. A Python analytics module (using Pandas and related tools) for data processing, exploratory analysis, and generating insights. 4. (Optional) An AI-powered natural language querying feature to allow users to ask questions and receive data-driven responses.
Project Success Criteria: The project will be considered successful if: • The system successfully integrates with the YourQS database (Firebird/MS Access) and retrieves relevant data. • The platform demonstrates effective end-to-end data flow between backend (C#), analytics layer (Python), and frontend (Blazor).

- Users can **filter, search, and analyse construction data**, including trends, averages, and comparisons.
- The system supports **“what-if” analysis**, such as comparing material choices (e.g., concrete vs timber).
- Interactive dashboards clearly present insights in a **user-friendly and intuitive interface**.

The system meets key non-functional requirements:

Performance: Efficient data processing and response times

Usability: Suitable for non-technical users (architects, estimators)

Reliability: Accurate and consistent analytical outputs

Security: Safe handling of sensitive cost data, especially with AI integration

Scalability & Maintainability: Modular architecture supporting future enhancements

- (If implemented) The AI component can interpret natural language queries and return relevant analytical insights.

Appendix D: Work Breakdown Structure (WBS)

Prepared by: Becky Zhou

Date: 22/03/2026

1.0 Planning Phase

1.1 Project Initiation

- 1.1.1 Kick-start meeting
- 1.1.2 Team contract
- 1.1.3 Identify team roles and responsibilities
- 1.1.4 Set up shared workspace and communication tools

1.2 Project Scope & Rationale

- 1.2.1 Define project rationale and goals
- 1.2.2 Define project objectives and scope
- 1.2.3 Conduct client meeting to confirm scope

1.3 Develop Project Plans

- 1.3.1 Work Breakdown Structure (WBS)
- 1.3.2 Project schedule (Sprint roadmap)
- 1.3.3 Risk register
- 1.3.4 Issue log
- 1.3.5 Communication management plan
- 1.3.6 Change management plan
- 1.3.7 Quality assurance (QA) plan
- 1.3.8 Feasibility assessment (skills & tools)
- 1.3.9 Skills analysis
- 1.3.10 Upskilling plan
- 1.3.11 Stakeholder register

1.4 Proposal Development

- 1.4.1 Compile proposal document
- 1.4.2 Internal review and QA
- 1.4.3 Mentor review and feedback
- 1.4.4 Revise proposal
- 1.4.5 Finalise and proofread

1.5 Proposal Submission & Presentation

[Milestone] Proposal approved

Deliverable: Approved proposal (10%)

2.0 Learning Phase

- 2.1 Identify required skills and tools
- 2.2 Data science and analytics upskilling
- 2.3 Agile/Scrum upskilling
- 2.4 Tool and environment setup
 - 2.4.1 Install development tools (Python, R)
 - 2.4.2 Set up GitHub
 - 2.4.3 Set up project management tools
 - 2.4.4 Test dataset access

2.5 Learning review and documentation

[Milestone] Mid-project review completed

Deliverable: Status Report

3.0 Analysis Phase

- 3.1 Understand requirements (Sprint 1)
- 3.2 Data understanding and exploration
- 3.3 Perform exploratory data analysis (EDA)
- 3.4 Identify key metrics (cost per m², trade costs)
- 3.5 Define dashboard requirements

4.0 Design Phase

- 4.1 Design system architecture
- 4.2 Design dashboard layout and visualisations
- 4.3 Define data flow and integration

5.0 Implementation Phase

- 5.1 Develop web dashboard
- 5.2 Implement API integration
- 5.3 Develop data visualisations
- 5.4 Integrate system components
- 5.5 Implement advanced features (what-if analysis / AI queries)

6.0 Testing Phase

- 6.1 Functional testing
- 6.2 Usability testing
- 6.3 Bug fixing and improvements
- [Milestone]** System tested and validated

7.0 Deployment Phase

- 7.1 Deploy system
- 7.2 Documentation
- 7.3 Client handover
- 7.4 Client Feedback
 - 7.4.1 Prepare feedback form
 - 7.4.2 Collect feedback
- Deliverable: Client feedback
- 7.5 Final Portfolio & Product
 - 7.5.1 Finalise artefacts
 - 7.5.2 Organise portfolio
- [Milestone]** Portfolio submitted
- Deliverable: Portfolio
- 7.6 Reflective Report (Individual Assessment)
 - 7.6.1 Reflect on project
 - 7.6.2 Draft and revise

Deliverable: Reflective report (10%)

- 7.7 Poster Presentation (Assessment)
 - 7.7.1 Design poster
 - 7.7.2 Prepare presentation
 - 7.7.3 Present
- [Milestone]** Final presentation completed
- Deliverable: Poster

8.0 Project Management (Agile Scrum, Ongoing)

8.1 Sprint planning (2-week cycles)

8.2 Weekly team meetings

8.3 Progress tracking and issue management

8.4 Worklog documentation

Appendix E: Milestone Report

Prepared by: Becky Zhou

Date: 26/03/2026

Milestone	Date	Status	Responsible	Issues/Comments
Kick-start meeting completed	Week 1	Completed	Team	Initial understanding established
Project scope and objectives confirmed	Week 2	Completed	Team + Client	Minor clarification needed
Project plans completed (WBS, risk, etc.)	Week 3-4	Completed	Team	No major issues
Proposal submission	Week 5	Completed	Team	Submitted on time
Proposal presentation	Week 6	Pending	Team	Awaiting feedback
Proposal approval	Week 6	Pending	Supervisor + Moderator	Required to proceed
Learning and upskilling completed	Week 7	Pending	Team	Some tools need more practice
Initial data exploration completed	Week 8	Pending	Data lead	Possible data quality issues
Prototype dashboard (basic version)	Week 10	Pending	Dev team	Time constraint risk
Mid-project progress review	Week 12-13	Pending	Team + Supervisor	Critical milestone
Review feedback and update plan	Week 13	Pending	Team	Adjust scope if needed
System refinement and design updates	Week 14	Pending	Dev team	Based on feedback
Continued development (Sprint cycles)	Week 15-17	Pending	Dev team	Iterative improvements
Advanced features implementation	Week 18-19	Pending	Dev team	Depends on complexity
Internal testing and validation	Week 20-21	Pending	Team	Ensure enough testing time
Client feedback collection	Week 24	Pending	Team + Client	Important for grading
Final portfolio submission	Week 25	Pending	Team	Includes all artefacts
Reflective report submission	Week 25	Pending	Individual	Individual task
Poster presentation	Week 25-26	Pending	Team	Final showcase
Project handover to client	Week 26	Pending	Team + Client	Final delivery
Kick-start meeting completed	Week 1	Completed	Team	Initial understanding established

Project scope and objectives confirmed	Week 2	Completed	Team + Client	Minor clarification needed
Project plans completed (WBS, risk, etc.)	Week 3–4	Completed	Team	No major issues
Proposal submission	Week 5	Completed	Team	Submitted on time
Proposal presentation	Week 6	Pending	Team	Awaiting feedback
Proposal approval	Week 6	Pending	Supervisor	Required to proceed

Appendix F: Risk Register

Risk Register for Analysis and Presentation of Construction Data

Prepared by: Jackson Pull			Date: 22/03/2026								
No.	Rank	Risk	Description	Category	Root Cause	Triggers	Potential Responses	Risk Owner	Probability	Impact	Status
1	1	Difficulty integrating with the Firebird database	Issues connecting to or retrieving data from the database	Technical	Complex or outdated database structure	Connection errors, failed queries	Early testing, step-by-step query development, documentation review	Team			Open
2	2	Time constraints impacting project completion	Limited time to complete all required features	Project Management	Tight deadlines or underestimation of workload	Delays in milestones or incomplete features	Prioritise core features, adjust scope, track progress regularly	Team			Open
3	3	Performance issues with large datasets	Slow processing or system lag when handling large data	Technical	Inefficient queries or large data volume	Long load times, system slowdowns	Optimise queries, limit data, use indexing	Team			Open
4	4	Limited experience with data analytics tools	Lack of familiarity with tools	Resource	Limited prior experience	Development delays or errors in analysis	Research, tutorials, practice, seek support	Team			Open
5	5	Incorrect interpretation of data insights	Misunderstanding or misrepresenting analysis results	Data	Lack of domain knowledge or unclear outputs	Inconsistent or misleading conclusions	Validate results, review outputs, simplify visualisations	Team			Open
6	6	Communication issues within the team or with the client	Unauthorised access to sensitive cost data	Security	Weak access controls or poor data handling	Data breaches or unauthorised access	Implement authentication, restrict permissions	Team			Open
7	7	Communication issues within the team or with the client	Poor communication leading to misunderstandings	Project Management	Lack of regular updates or unclear expectations	Confusion, delays, or rework	Regular meetings, use Teams/OneDrive, clear documentation	Team			Open
8	8	Technical issues with system development or deployment	Errors during development or when deploying the system	Technical	System complexity or integration challenges	Build failures, runtime errors	Testing, debugging, version control	Team			Open
9	9	Challenges in designing effective data visualisations	Visual outputs are unclear or difficult to interpret	Usability	Poor design choices or too much information	User confusion or misinterpretation	Keep visuals simple, gather feedback, iterate design	Team			Open
10	10	Fluctuations in material and construction costs affecting data accuracy	Changes in market prices reduce reliability of analysis	External	Market variability and inflation	Outdated or inconsistent cost data	Use recent data, update datasets regularly	Team			Open

Appendix G: Issue Log Register

List of Prioritized Risks for Analysis and Presentation of Construction Data

Prepared by: Jackson Full

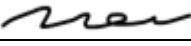
Date: 20/03/2026

Ranking	Potential Risk
1	Difficulty integrating with the Firebird database
2	Time constraints impacting project completion
3	Performance issues with large datasets
4	Limited experience with data analytics tools
5	Incorrect interpretation of data insights
6	Data security and access control issues
7	Communication issues within the team or with the client
8	Technical issues with system development or deployment
9	Challenges in designing effective data visualisations
10	Fluctuations in material and construction costs affecting data accuracy

Appendix H; Team Contract

Project Name: Analysis and Presentation of Construction Data

Project Team Members Names and Sign-off:

Name	Sign-off on Team Contract
Becky	
Jackson	
Max	
Mantej	

Code of Conduct: As a project team, we will:

- Act professionally and treat all team members with respect.
- Share relevant project information with the whole team.
- Stay committed to the project and complete our responsibilities to the end.

Participation: We will:

- Contribute fairly and honestly to all team tasks.
- Respect different ideas and encourage everyone to participate.
- Be open to feedback and new ways of working.
- Inform the team early if we cannot attend a meeting or meet a deadline.

Communication: We will:

- Use agreed communication channels for updates and discussions.
- Communicate clearly, respectfully, and in a timely manner.
- Respond to messages and shared documents within a reasonable time.

Problem Solving: We will:

- Work together to address problems as a team.
- Listen to each other's ideas and build on possible solutions.
- Raise ongoing concerns with the mentor or lecturer if needed.

Meeting Guidelines: We will:

- Attend the regular weekly meeting every Thursday.
- Schedule any extra meetings in advance where possible.
- Assign responsibility for minutes, action items, and follow-up tasks.

Appendix I: Team Skills Register

Skill Area	Description	Current Level
C# / .NET	Backend development and integration	Beginner (5/20)
Blazor Frontend	Web UI development using C# and .NET	Beginner (5/20)
SQL	Querying, filtering, joins, aggregations	Intermediate (13/20)
Data Analysis	Data processing, analytics logic, data visualisation	Advanced (18/20)
Version Control (GitHub)	Code collaboration and version tracking	Advanced (17/20)
MS Access	Client data source	Beginner (3 / 20)

Note: The skill levels presented in the table above are based on an [internal team skills matrix](#).

Appendix J: Project Feasibility

Detailed Technologies:

Development technologies:

C# / .NET | Blazor | Python | SQL;

Data and storage:

Microsoft Access | DBeaver

Collaboration and project tools:

GitHub | OneDrive | Microsoft Teams | Lucidchart & Visual Paradigm | Visual Studio Code | Discord

Alternative Approaches:

Full Python-based platform

This approach would use Python for both backend and analytics. While efficient for data processing it doesn't align with the client's existing environment.

Full C# / .NET platform

This approach aligns with the client's infrastructure but was considered less feasible due to limited team experience and reduced efficiency for analytical tasks.

Hybrid development process (**SELECTED**)

This approach combines C# / .NET for backend integration with Python for analytics balancing feasibility and performance.

Feasibility Challenges:

The project includes following technical challenges:

Limited experience with C# / .NET and Blazor.

Time constraints due to academic workload.

Potential complexity of the Microsoft Access database structure.

Integration between different technologies (C# / .NET, Python and MS Access).

Data Considerations:

We will use a Microsoft Access database provided by client. Data will be explored, validated and potentially transformed into more suitable and comfortable formats (CSV, SQL-based systems) to support efficient analysis.

Appendix K: Quality Assurance Plan

Quality Assurance Plan for Analysis and Presentation of Construction Data

Prepared by: Jackson Full

Date: 27/03/2026

1.0 Draft Quality Assurance Plan

1.1 Introduction

This Quality Assurance Plan outlines the processes and standards used to ensure the quality of the prototype system developed for analysing and presenting construction data for YourQS. The plan focuses on maintaining accuracy, reliability, and usability throughout the project lifecycle.

1.2 Purpose

The purpose of this plan is to ensure that all project deliverables meet the required quality standards, function correctly, and align with client expectations. It also aims to minimise errors in data processing, system functionality, and outputs.

1.3 Policy Statement

The project team is committed to delivering a high-quality prototype that meets functional requirements and maintains data integrity. All work will follow defined quality assurance procedures, including testing, reviews, and validation against requirements.

1.4 Scope

This Quality Assurance Plan applies to all aspects of the project, including backend data processing, integration with the YourQS Firebird database, development of the web-based dashboard, generation of data visualisations, and all supporting documentation. It ensures that both technical components and deliverables are subject to quality control processes.

2.0 Management

2.1 Organizational Structure

The project team consists of student developers working collaboratively, with responsibilities distributed across project management, system development, and data analysis. The structure supports coordination, accountability, and effective quality control throughout the project.

2.2 Roles and Responsibilities

2.2.1 Technical Monitor/Senior Management

The client (YourQS) acts as the technical monitor by providing requirements, feedback, and approval of key project decisions. They ensure that the system aligns with real-world expectations and business needs.

2.2.2 Task Leader

The project manager is responsible for overseeing progress, ensuring that quality standards are met, managing risks, and coordinating communication between team members and stakeholders.

2.2.3 Quality Assurance Team

The quality assurance function is carried out by team members through structured reviews and testing. This includes verifying system functionality, validating data outputs, and ensuring compliance with project requirements.

2.2.3 Technical Staff

Technical team members are responsible for developing the backend and frontend components, implementing data processing and visualisation features, and resolving any defects identified during testing.

3.0 Required Documentation

- Scope statement
- Risk register
- Work Breakdown Structure (WBS)
- Project schedule
- Technical concept document
- Data analysis reports
- Final project report and presentation

4.0 Quality Assurance Procedures

4.1 Walkthrough Procedure

Regular walkthroughs will be conducted where team members present their work for review and feedback. These sessions help identify issues early in the development process and ensure alignment with project requirements.

4.2 Review Process

4.2.1 Review Procedures

A structured review process will be used to maintain quality across all deliverables. This includes peer reviews of code, validation of data analysis results, and evaluation of visualisations to ensure accuracy, clarity, and relevance to user needs.

4.3 Audit Process

4.3.1 Audit Procedures

Periodic audits will be carried out to verify that the system meets its functional and technical requirements. This includes checking database queries, validating system outputs, and ensuring documentation is complete and consistent.

4.4 Evaluation Process

The system will be evaluated based on its ability to produce accurate data outputs, perform efficiently, and provide a usable interface. Evaluation will also consider how well the system meets the project objectives and client expectations.

4.5 Process Improvement

Feedback from testing, reviews, and client input will be used to continuously improve both the system and the development process. Identified issues will be addressed promptly to enhance overall quality.

5.0 Problem Reporting Procedures

5.1 Noncompliance Reporting Procedures

Any issues such as system errors, incorrect data outputs, or failures to meet requirements will be documented and tracked. These issues will be prioritised and resolved to ensure that the final system meets quality standards.

6.0 Quality Assurance Metrics

Quality will be measured using metrics such as the accuracy of data outputs, number of system errors, system performance, completion of deliverables on time, and overall client satisfaction. These metrics will help assess whether the project meets its quality objectives.

Appendix

Quality Assurance Check List Forms

1. Data Accuracy Checklist

- Data is retrieved correctly from the YourQS database
- Queries return expected results
- Calculations (costs, averages, comparisons) are accurate
- No missing or duplicated data in outputs
- Results are consistent across multiple tests

2. System Functionality Checklist

- All core features are working (filtering, visualisation, comparison)
- “What-if” analysis functions correctly
- System integrates successfully with the database
- No critical errors or crashes during use
- All user inputs return valid outputs

3. User Interface Checklist

- Interface is clear and easy to use
- Visualisations (charts, graphs) are readable and accurate
- Navigation between features is smooth
- Filters and controls work as expected
- Results are displayed clearly

4. Performance Checklist

- System responds within acceptable time
- Large datasets are handled without major delays
- No significant lag when loading data or visuals

5. Security and Data Handling Checklist

- Data access is restricted to authorised users
- No sensitive data is exposed unnecessarily
- Data is not shared with external systems without approval
- Secure connection to database is maintained

6. Documentation Checklist

- All required documents are completed
- Documentation is clear and consistent
- Technical details are accurate
- Final report reflects actual system functionality

Appendix L: Cost Estimation

YourQS Ltd Project Cost Estimate

Prepared by: Mantej Singh

Date: 26/03/2026

Note: Change the WBS items and other entries to meet your project needs. This data is from Figure 7-1 of Schwalbe's text Information Technology Project Management, Fourth Edition. Also make sure the formulas work properly based on the data you enter.

WBS Items	# Number of unit/month/hour	Cost	Subtotals
1. Project Management			
1.1 Project manager Cost per Hour	24	\$163.30	\$3,919.20
1.2 Project team members Cost per Hour	1440	\$30	\$43,200
Miscellaneous Cost			
Internet Costs			\$200
Transport Costs (week)	24	\$80	\$1,920
2. Hardware			
2.1 Laptops (units)	5	\$1,000	\$5,000
2.2 Server (months)	9	\$50	\$450
3. Software			
3.1 AI integration Licensed software(months)	9	\$100	\$900
5. Training and Support	9 months	\$23	\$207
Youtube Subscription Fees (monthly)			
6. Reserves (10)% of total estimate			\$5,579.6
Total project cost estimate			61,375.8

1)Project Team member Cost: The Team labour cost is based on the approximate number of hours each group member will be spending on the project over the course of two semesters (24 weeks). The total hours come to 1440, there with an hourly rate of \$30 the final figure of \$43,200 is applied. The hourly rate of \$30 is based on the typical entry level IT job in the New Zealand Industry. This makes sure that the estimates along with the industry figures give us a realistic idea of the cost.

2)Mentor Manager Cost: The mentor cost is based on the workshop guidance where it states that a rate of \$142 plus GST (15%) is applied for every meeting. This means that assuming we have a meeting every week with our mentor over a course of roughly 24 weeks we get \$3913 (24 X \$163.30).

3)Miscellaneous Costs: These costs include any expenses such as internet usage, cloud usage, software, transport etc: Based on the meeting and requirements from our client we will be heavily working on the Firebird Database, Python and SQL based tools. Based on the Canstar blue (NZ) average broadband range from \$70 - \$100, a cost of \$200 has been included for any shared plan etc:

Along with this transport cost plays a crucial role, since we will be meeting twice a week with the whole group in university. Based on the Auckland transport and parking costs assuming \$20 for each person a week, we get a total of \$1,920 over a course of 24 weeks.

4) Hardware and Software Costs: These costs include things like laptops, servers, AI software and Training and support expenses. Following costs are vital to consider as they reflect a realistic real-world idea of the project. For example, laptops are already available to the team members, but the costs are stated to represent the true cost of delivering the solution in an environment. Along with this other Ai software and training support expenses such as YouTube premium are included to reflect on the future scalability and advancement in the project.

5) Contingency Reserve (10%): This cost is applied to the subtotal of \$47,319 to account for any uncertainties that might appear during the course of this project. This includes working with client databases, paying for new software, data quality issues, increased development time etc: Therefore, in order to manage these risks a reserve amount is included.

6) Scope of the project: The cost estimations above reflect the development cost of the project by the project team only. This means that any other cost forms the client side such as staff time, infrastructure, operating cost or not included. This is done as the project is focused on delivering the final project prototype rather than a full production system.

Appendix M: Detailed Project Schedule – Gantt Chart

